UNIVERSITY OF THE PUNJAB

Roll No. ....

Fifth Semester 2017

Examination: B.S. 4 Years Programme

PAPER: Design and Analysis of Experiments (Theory)
Course Code: STAT-303

TIME ALLOWED: 30 mins.
MAX. MARKS: 10

Attempt this Paper on this Question Sheet only.

OBJECTIVE TYPE

Q1. Read the following items carefully and encircle the correct option listed below at each item.
(One mark for each)

i) **The basic principles of experimental designs consist of:**
a) Randomization b) Replication c) Local Control d) All of these

ii) **The smallest subdivision of the experimental material is called:**
a) Treatments b) Experimental Unit c) Experimental Error d) None of these

iii) **In a completely randomized design, treatments are assigned to experimental units .... at random.**
a) Completely b) Partially c) Systematically d) None of these

iv) **The assumptions under analysis of variance consist of:**
a) Normality and Independence
b) Both (a) and (b)
c) Linearity and Additivity
d) None of these

v) **The following design provides the maximum number of degrees of freedom for error sum of squares:**
a) Completely Randomized Design
b) Latin Square Design
c) Completely Randomized Block Design
d) None of these

vi) **Multiple comparisons tests are applicable when:**
a) Null Hypothesis about equality of means is rejected
b) Null Hypothesis about equality of means is accepted
c) Does not depend upon the rejection or acceptance of Null Hypothesis
d) None of these

vii) **One can estimate the missing observation through covariance technique by simply changing the sign of....**
a) b b) r c) Correction Factor d) None of these

viii) **The efficiency of two experimental designs can simply be measured through ..... of error variances.**
a) Addition b) Subtraction c) Multiplication d) Ratio

ix) **Two Latin squares are .... if each letter of one square design occurs exactly once with every letter of the other square when they are superimposed.**
a) Orthogonal b) Factorial Designs c) Efficient d) None of these

x) **A contrast is ....combination of treatments.**
a) Linear b) Exponential c) Quadratic d) None of these

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Fifth Semester 2017
Examination: B.S. 4 Years Programme

Roll No. ....

PAPER: Design and Analysis of Experiments (Theory) **TIME ALLOWED: 2 hrs. & 30 mins.**
Course Code: STAT-303 **MAX. MARKS: 50**

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE TYPE

SHORT QUESTIONS

Q2. Explain the following: *(6,3,4,3,4)*

i) Random Effects, Fixed Effects and Mixed Effects Models
ii) Analysis of Covariance
iii) Principles of Experimental Design
iv) Assumptions under Analysis of Variance
v) Orthogonal Polynomials.

SUBJECTIVE

Q3. a) In an experiment 'k' treatments and 'r' blocks are selected at random from a large number of treatments and blocks. Develop expected mean squares by clearly indicating the assumptions used.

b) Given the following ANOVA for a CR design for four treatments: (6+4)

S.O.V	d.f.	SS
Treatments	3	1.1986
Error	36	1.0323

Test the significance of difference between treatment means by using Duncan's Multiple Range Test when treatment means for four treatments were 1.464, 1.195, 1.325, and 1.66.

Q4. a) Seven treatments arranged in six randomized complete blocks gave the following sum of squares and products

S.O.V.	YY	XY	XX
Blocks	1200	600	200
Treatments	800	300	100
Error	1400	700	600

i) Is the regression of Y on X significant at 0.05 level of significance?
ii) Construct ANOVA and write the inference.

b) The analysis of Variance for a RCB design produced the table shown below: (6+4)

S.O.V	d.f.	SS	MS	F-Ratio
Treatments	3	28.2	-	
Blocks	5	-	13.80	
Error	-	34.1	-	

Complete the ANOVA table and test the significance of difference among the treatment means.

Q5. a) Derive formula for estimating N missing observations in a LS Design when values are missing in different rows, different columns and different treatments.

b) In an experiment to examine the effects of row spacing on the yield of wheat, 8 row spacing were used and 6 blocks of an experiment were used. The sum of squares for Total, Blocks and Treatments were 2195.48, 617.86 and 1283.65 respectively. Find the relative efficiency of this design with the design in which blocks are ignored. *(7+3)*